CLINICAL · NOSE & SINUSES

Proptosis and the orbit

The orbit shares thin walls with three sinuses, so the rhinologist meets its diseases often: the eye pushed out by a mucocele, a sinus infection or a tumour, the protruding eyes of thyroid disease that the nose can decompress, and the watering eye that a dacryocystorhinostomy can drain. Proptosis is a sign; its direction, speed and sidedness point to the cause.

AT A GLANCE

- Thyroid eye disease is the commonest cause in adults.
- Orbital cellulitis is the commonest cause in children.
- Direction of displacement points to the source.
- Retrobulbar haemorrhage: lateral canthotomy within minutes.
- Drain a subperiosteal abscess for age ≥9, large or non-medial abscess, frontal sinusitis, visual change or no improvement in 48 h.

1 CHAPTER ONE Overview

The eye pushed forward.

1.1 Definitions

Proptosis is forward displacement of the eyeball; *exophthalmos* is often used for proptosis from thyroid eye disease. Measured with an exophthalmometer (Hertel): a difference of more than 2 mm between the eyes, or a reading above about 20 mm, is abnormal. *Pseudoproptosis* (a large eye in myopia, lid retraction, or a small opposite eye) must be excluded.

1.2 Applied anatomy

The orbit is a four-sided pyramid. Its medial wall (the lamina papyracea) separates it from the ethmoids, its floor from the maxillary sinus, its roof from the frontal sinus and anterior cranial fossa; the sphenoid lies at the apex beside the optic canal. The

orbital periosteum and septum limit the spread of infection; the valveless ophthalmic veins drain to the cavernous sinus.

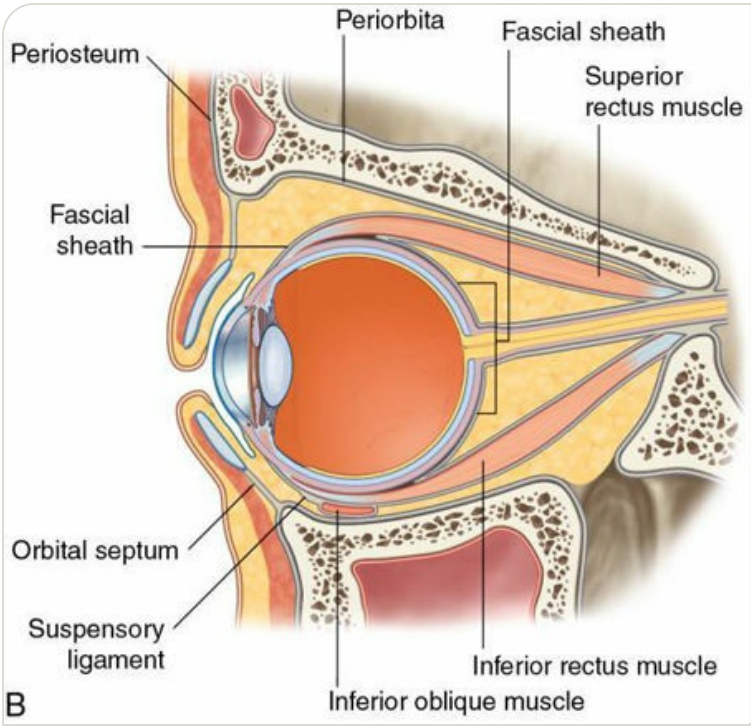


FIGURE 1 Anatomy of the orbit: its bones and relations.

Dhingra, fig. 41.1

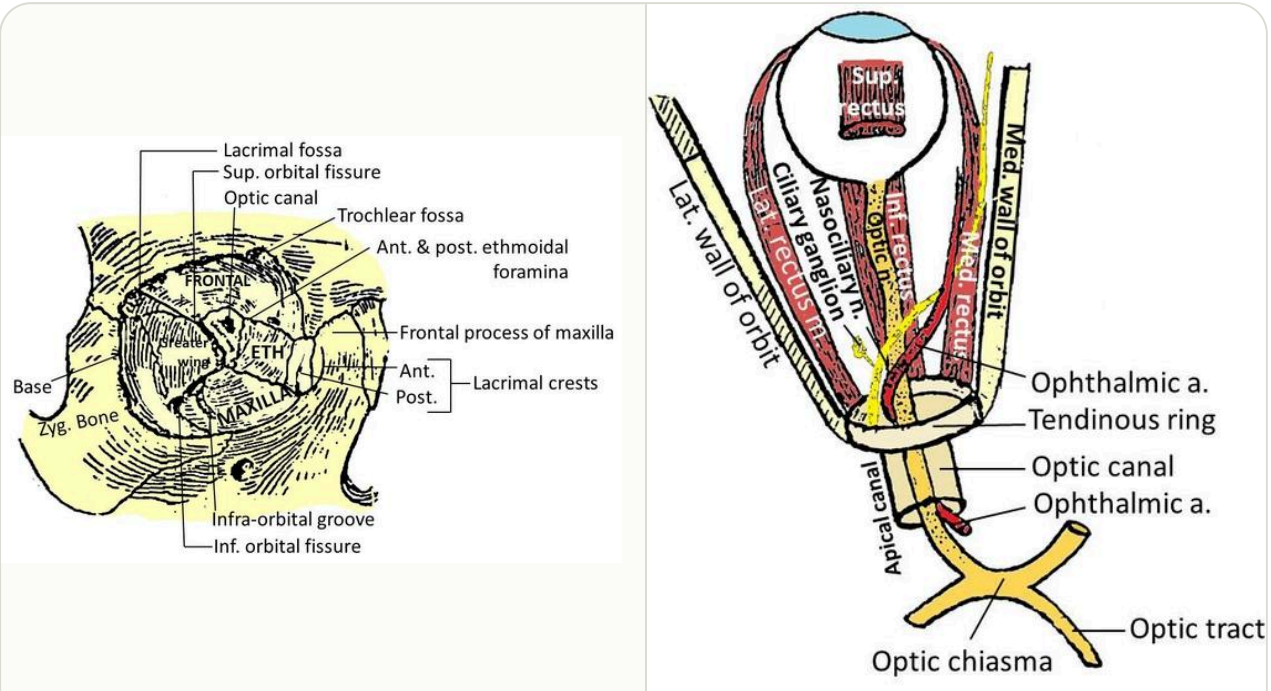


FIGURE 2 The bony orbit with its walls, fissures and foramina, and the optic canal and tendinous ring at the apex.

Anatomy atlas · Doss p. 90, Doss p. 95

1.3 Why it matters for ENT

Sinus infections, mucoceles, fungal disease, polyps, angiofibroma, inverted papilloma and sinonasal cancers all invade the orbit; endoscopic surgery can injure it; and the orbit can be decompressed and drained through the nose.

2 CHAPTER TWO Causes

VEIN.

CAUSES OF PROPTOSIS (VEIN)

GROUP	EXAMPLES
Vascular	Venous varix, cavernous haemangioma, carotid-cavernous fistula, orbital haematoma
Endocrine	Thyroid eye disease (Graves ophthalmopathy)
Inflammatory and infective	Orbital cellulitis and abscess, idiopathic orbital inflammation, fungal sinusitis, granulomatosis with polyangiitis, dacryoadenitis
Neoplastic	Primary orbital tumours, spread from the sinuses or cranium, metastases

2.1 From the sinuses

Mucoceles of the frontal and ethmoid sinuses, allergic fungal polyposis, inverted papilloma, angiofibroma, olfactory neuroblastoma and sinus carcinoma, and orbital complications of sinusitis (see those topics).



FIGURE 3 First, proptosis with lateral displacement of the eye from allergic fungal polyposis. Second, a left ethmoid mucocele displacing the eye.

Dhingra, fig. 41.2

2.2 Clues from the history and examination

- **Unilateral or bilateral:** thyroid eye disease is the commonest cause of both bilateral and unilateral proptosis in adults; orbital cellulitis is the commonest cause in children.
- **Speed:** sudden (haemorrhage, cellulitis), weeks (inflammation, malignancy), years (benign tumours, mucoceles, fibrous dysplasia).
- **Direction:** *axial* proptosis suggests a lesion inside the muscle cone (haemangioma, optic nerve tumour, thyroid eye disease); *non-axial* displacement points away from the lesion: down and out with a frontoethmoid mucocele, laterally with an ethmoid lesion, upwards with a maxillary tumour.
- **Variable proptosis** with straining or bending: venous varix.
- **Pulsatile proptosis** with a bruit and arterialised conjunctival vessels: carotid-cavernous fistula.
- **Pain:** inflammation, infection or malignancy.

3 CHAPTER THREE Important conditions

Inside and around the orbit.

3.1 Thyroid eye disease

The commonest cause of proptosis in adults, bilateral or one-sided, in hyperthyroid, euthyroid or hypothyroid patients. Lid retraction, lid lag, chemosis, restricted movement and exposure keratopathy; CT shows enlargement of the muscle bellies with sparing of their tendons. Optic nerve compression at the apex threatens vision.

3.2 Idiopathic orbital inflammation

Orbital pseudotumour: painful proptosis, worse on eye movement; may involve the muscles (myositis, including the tendons, unlike thyroid disease), lacrimal gland, sclera or optic nerve. Biopsy shows non-specific inflammation; treated with steroids, immunosuppression or radiotherapy.

3.3 Vascular lesions

- **Cavernous haemangioma:** the commonest benign orbital tumour of adults; painless, slowly progressive axial proptosis from a well-defined, enhancing intraconal mass; excised by lateral orbitotomy or endoscopically.
- **Capillary haemangioma:** the commonest orbital tumour of infants; usually involutes by about seven; propranolol or steroids.
- **Venous varix:** proptosis that comes and goes with straining.
- **Carotid-cavernous fistula:** pulsatile proptosis, bruit, chemosis and arterialised vessels; endovascular closure.

3.4 Tumours

- **Lymphoma:** the commonest malignant orbital tumour of adults, often anterior and palpable; biopsy, radiotherapy and chemotherapy.
- **Rhabdomyosarcoma:** the commonest primary orbital malignancy of children (around

six or seven years), with rapid proptosis; chemoradiation.

- **Dermoid cyst:** the commonest benign orbital tumour of children, at the frontozygomatic suture or deeper; excised.
- **Optic nerve glioma** (children, neurofibromatosis) and **optic nerve sheath meningioma** (adults).
- **Lacrimal gland tumours:** pleomorphic adenoma and adenoid cystic carcinoma in the upper outer orbit.
- **Metastases:** breast most often, then lung, prostate and kidney; neuroblastoma in children.

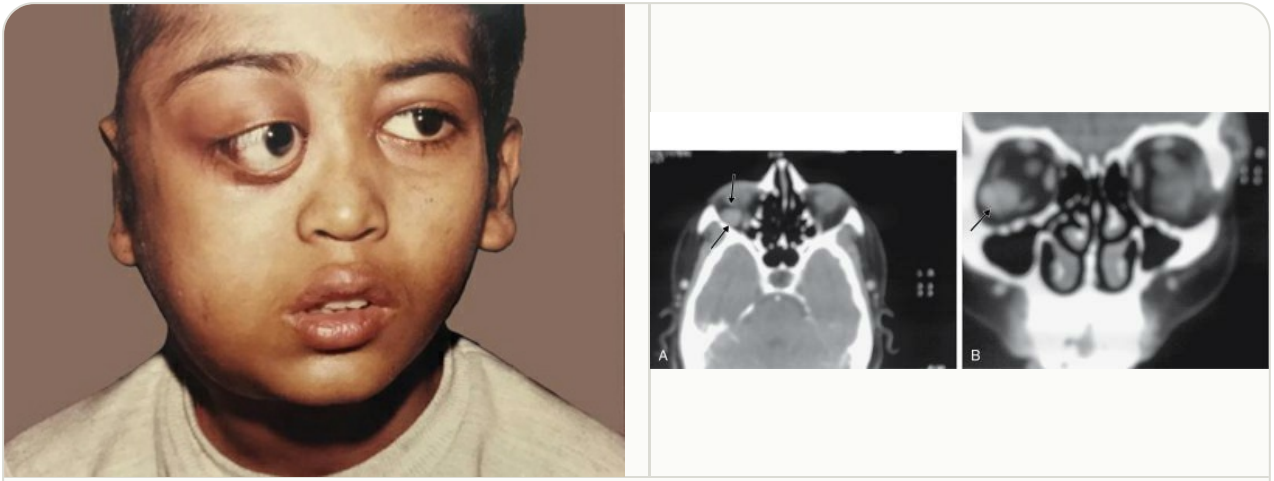
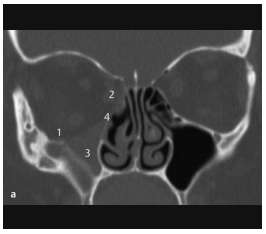


FIGURE 4 First, proptosis with medial displacement of the eye from lymphoma in a child. Second, a dermoid cyst of the right orbit on axial and coronal CT.

Dhingra, figs 41.4 and 41.5

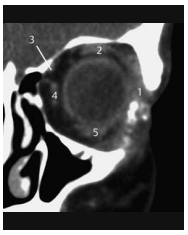
IMAGING IN THE APP 5 cases

Orbital decompression for Graves disease, orbital amyloidosis, mucocele, osteoma and orbital cellulitis.



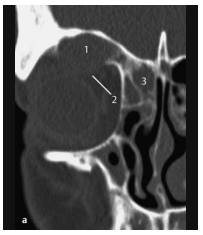
Orbital Decompression in Graves Disease

After surgery · 2 images



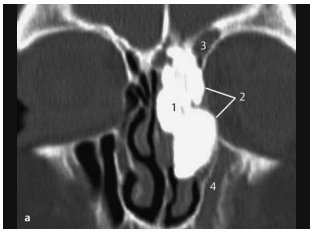
Amyloidosis in the Orbit

Infection & inflammation · 1 image



Mucocele

Infection & inflammation · 13 images



Osteoma, Gardner Syndrome, and Complications

Bone disease · 3 images



Orbital Cellulitis

Infection & inflammation · 6 images

4 CHAPTER FOUR Evaluation

A joint ENT and eye assessment.

4.1 Examination

Visual acuity, colour vision and pupils (a relative afferent pupillary defect means optic nerve compromise), eye movements and diplopia, the degree and direction of proptosis, lids, conjunctiva and cornea, fundus, and nasal endoscopy.

4.2 Investigations

- **CT** of the orbits and sinuses: bone, sinus disease, fractures, muscle size.
- **MRI** for soft tissue, the optic nerve, apex and intracranial extension.
- Thyroid function and antibodies; blood tests for vasculitis and lymphoma as indicated.
- Biopsy for undiagnosed masses, endoscopically or by orbitotomy.

5 CHAPTER FIVE Procedures

What the nose can do for the eye.

5.1 Endoscopic orbital decompression

Removal of the medial wall (lamina papyracea) and the medial part of the floor, opening the periorbita so orbital fat can prolapse into the ethmoid and maxillary spaces. For thyroid eye disease with optic neuropathy, severe exposure or disfiguring proptosis. Diplopia may follow.

5.2 Optic nerve decompression

Removal of the bone of the optic canal through the sphenoid in traumatic or compressive optic neuropathy; its value in trauma is debated.

5.3 Dacryocystorhinostomy

For epiphora from obstruction of the nasolacrimal duct: a window is made from the lacrimal sac into the middle meatus, through the lacrimal bone and frontal process of the maxilla, endoscopically or through an external skin incision, sometimes with silicone tubes.

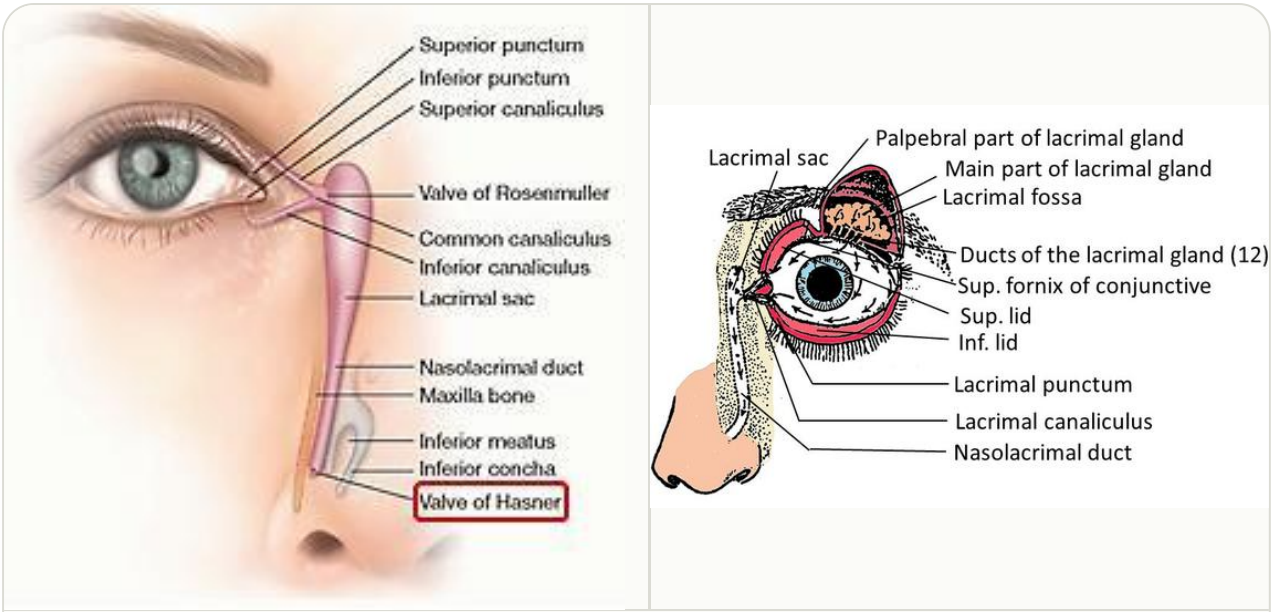


FIGURE 5 The lacrimal drainage system, and the lacrimal apparatus.
Anatomy atlas · ENT Steps p. 100, Doss p. 92

5.4 Drainage of orbital abscess

Endoscopic or external drainage of a subperiosteal or orbital abscess complicating sinusitis.

5.5 Lateral canthotomy and cantholysis

An emergency release for retrobulbar haemorrhage after trauma or sinus surgery: a tense, proptosed eye with rising pressure and falling vision. Cut the lateral canthal tendon and its inferior crus within minutes.

RETROBULBAR HAEMORRHAGE

Pain, tense proptosis, a hard globe, ophthalmoplegia and loss of vision after injury or sinus surgery. Do not wait for imaging: perform lateral canthotomy and cantholysis at once, then call ophthalmology.

5.6 Orbital exenteration

Removal of the orbital contents for malignancy invading beyond the periorbita.

OPERATIVE GUIDES IN THEATRE 6 operations

Orbital Decompression
Orbital decompression
Optic Nerve Decompression
Optic nerve decompression
Dacryocystorhinostomy
Dacryocystorhinostomy
Orbital Abscess
Orbital abscess drainage
Lateral Canthotomy
Lateral canthotomy
Orbital Exenteration
Orbital exenteration

WHO NEEDS DRAINAGE IN CHILDREN

Most children with orbital complications are treated medically at first, with intravenous antibiotics covering *Streptococcus anginosus* (milleri), the commonest organism in complicated sinusitis in young people, plus nasal decongestant, and with vision, pupils, colour vision and eye movements checked at least every few hours. A small **medial subperiosteal abscess** in a child with normal vision often resolves without surgery. Features that favour early drainage (based on the criteria of Garcia and Harris) are age nine years or older, frontal sinusitis, a non-medial or large abscess, gas in the abscess, a dental source, any reduction in visual acuity or colour vision, an afferent pupillary defect, intracranial complications, or failure to improve within 24 to 48 hours. Adolescents with frontal sinusitis and orbital involvement need imaging of the brain as well, because *S. anginosus* readily forms intracranial abscesses and Pott's puffy tumour.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Causes of proptosis** (VEIN) and the commonest in adults and children.
- **Direction of displacement** as a clue.
- **Thyroid eye disease** and endoscopic orbital decompression.
- **Commonest orbital tumours** in adults and children.

- **Retrobulbar haemorrhage** and lateral canthotomy.
- **Dacryocystorhinostomy.**

6.2 Common pitfalls

COMMON PITFALLS

- Waiting for a scan in retrobulbar haemorrhage.
- Missing a sinus cause in one-sided proptosis.
- Forgetting colour vision and the pupils when assessing the optic nerve.

Written for this app from Oxford Textbook of Otolaryngology ch. 36, Oxford Specialist Handbook ch. 21, Oxford Handbook of ENT ch. 9, Dhingra ch. 41 and K. J. Lee ch. 58. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

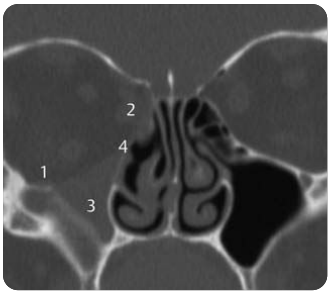
On imaging

The imaging of this topic: key findings, differentials and representative scans.

ORBIT & SINUS RELATIONS · AFTER SURGERY

Orbital Decompression in Graves Disease

- Conventional sinonasal radiography in these patients is not sufficient for the evaluation of concomitant sinusitis. CT will be needed for anatomic orientation, especially if endoscopic surgery is contemplated.
- Endonasal surgical procedures must be done with care due to distortion of the normal anatomy and an increased risk of damage to displaced intraorbital structures such as the optic nerve.

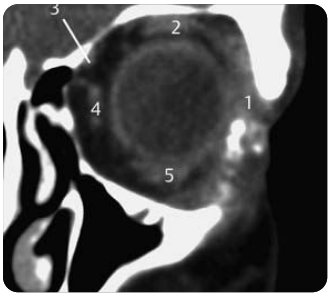


CT, coronal. The orbital floor (1) and the lamina papyracea (2) have been completely removed. Opacification of the maxillary sinus (3) may be due to…

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Amyloidosis in the Orbit

- Differentiation between solitary involvement of the orbit and involvement or growth from adjacent structures.
- Differentiation between unilateral and bilateral disease.
- Signs of inflammation.



CT, coronal. On this soft-tissue setting, a diffuse infiltrating lesion, also encasing and infiltrating the lateral rectus muscle, is present in th…

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Mucocele

IMAGING FINDINGS

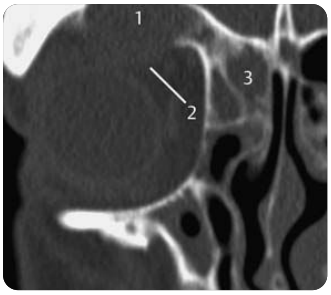
- Expanded, airless sinus with smooth remodelled walls
- Variable density and signal with protein content
- Rim enhancement only

TOP DIFFERENTIALS

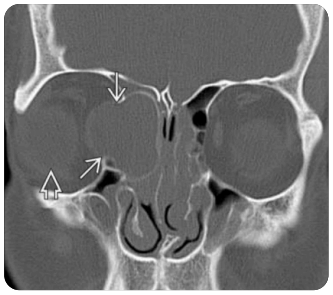
- Allergic fungal sinusitis
- Polyposis
- Tumour (solid enhancement, destruction)

CHECKLIST

- Orbit and skull base erosion
- Frontal recess obstruction as the cause



CT, coronal. The left frontal sinus is opacified with expansion of its inferior bony outline through the roof of the orbit (1). Although this is a bone-…



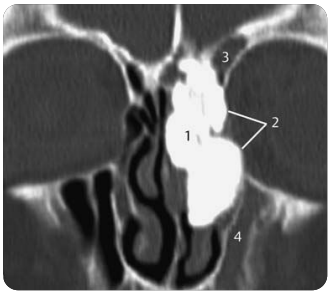
Coronal bone CT shows the typical features of a right ethmoid mucocele. The sinus is opacified, and there is remodeling of the surrounding bony…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716.

ORBIT & SINUS RELATIONS · BONE DISEASE

Osteoma, Gardner Syndrome, and Complications

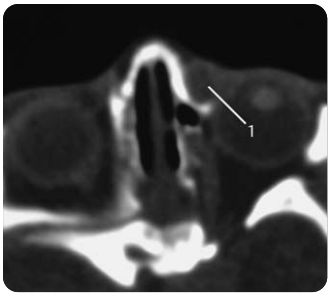
- Extension to structures which, at surgical removal, may result in cosmetic or functional complications.
- In Gardner syndrome, multiple osteomas may be found as well as other pathologies, some of which are at an increased risk of malignancy (colorectal carcinoma and thyroid carcinoma).



CT, coronal, bone-tissue setting. Note the large osteoma (1) protruding through the lamina papyracea (2) and obstructing the frontal recess (3)….

ORBIT & SINUS RELATIONS · CONGENITAL

Nasolacrimal Mucocele



Patient with frequent and fluctuating subcutaneous swellings below the left medial canthus.

Dunnebier, Imaging for…

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Orbital Cellulitis

IMAGING FINDINGS

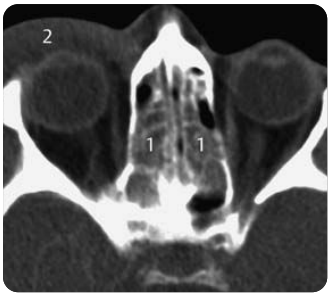
- Stranding of the postseptal fat
- Swollen extraocular muscles; proptosis
- No discrete collection

TOP DIFFERENTIALS

- Subperiosteal abscess
- Invasive fungal infection
- Idiopathic orbital inflammation

CHECKLIST

- Separate preseptal from postseptal
- Repeat imaging if not improving



CT, axial with intravenous contrast. Bilateral opacification of the ethmoid (1). The lamina papyracea is intact without signs of periosteal…



Axial CECT in a patient with facial impetigo shows edema and enhancement of the preseptal soft tissues with normal appearance of th…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 814.